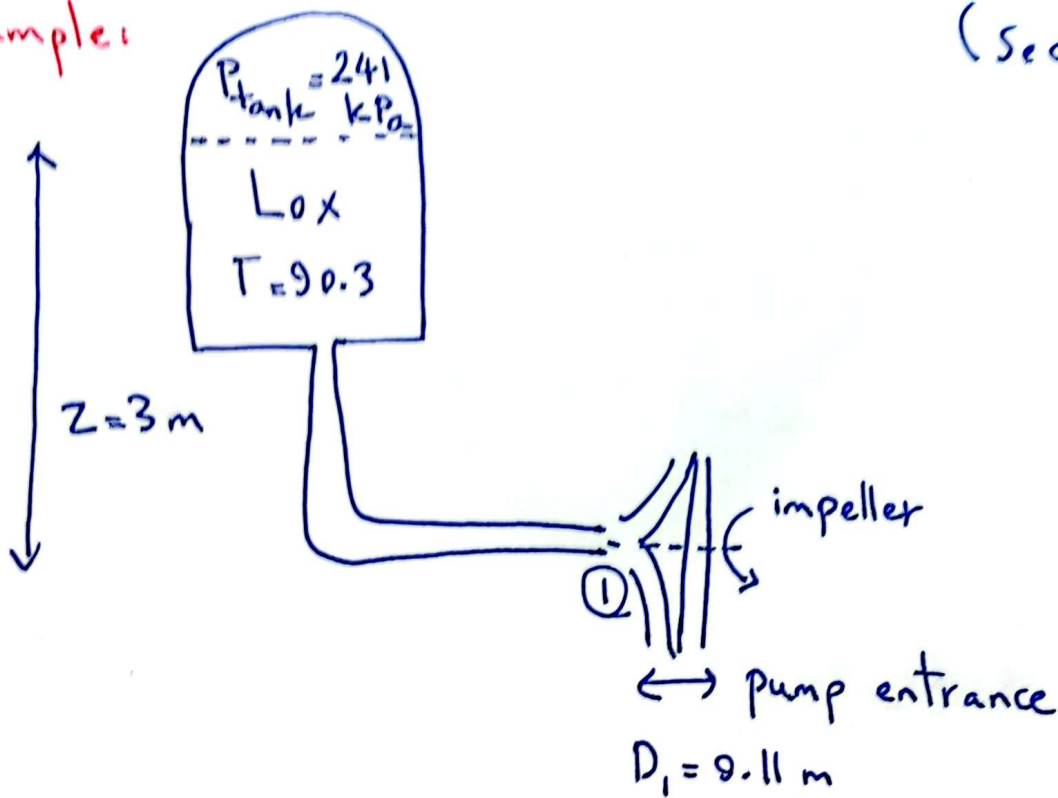


Example:

(Section 4)



$$SG_{\text{Lox}} = 1.14 \rightarrow \rho_{\text{Lox}} = 1140 \quad \dot{m}_1 = 182 \frac{\text{kg}}{\text{s}}$$

$z_{\text{loss}} = 1\text{m}$ (viscous force) check if cavitation occurs in pump or no?

Solution:

$$P_{o1} = \underbrace{P_{\text{tank}}}_{\text{pushing pressure}} + \underbrace{\rho_{\text{Lox}} g z}_{\text{potential energy}} - \underbrace{\rho_{\text{Lox}} g z_{\text{loss}}}_{\text{frictional loss}}$$

We are interested in equivalent head, so divide by $\rho_{\text{Lox}} g$

$$\Rightarrow \frac{P_{o1}}{\rho_{\text{Lox}} g} = \frac{P_{\text{tank}}}{\rho_{\text{Lox}} g} + \underbrace{z}_{\text{potential head (P-4)}} - \underbrace{z_{\text{loss}}}_{\text{frictional head (P-1)}}$$

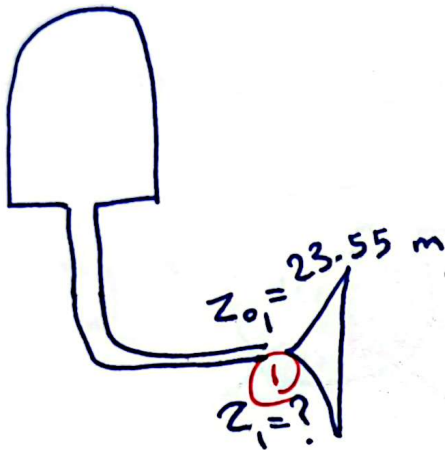
$\underbrace{\frac{P_{o1}}{\rho_{\text{Lox}} g}}_{Z_{o1}}$
 $\underbrace{\frac{P_{\text{tank}}}{\rho_{\text{Lox}} g}}_{\text{pressure head}}$

$$Z_{o_1} = Z_{\text{tank}} + Z - Z_{\text{loss}}$$

$$Z_{\text{tank}} = \frac{P_{\text{tank}}}{\rho_{\text{Lox}} g} = \frac{241 \times 10^3}{1140 \times 9.81} = 21.55 \text{ m}$$

$$\Rightarrow Z_{o_1} = 21.55 + 3 - 1 = 23.55 \text{ total head}$$

Now, let's compute Z_1 (static head), use Bernoulli Eqⁿ:



$$P_{o_1} = P_1 + \rho \frac{u_1^2}{2} \Rightarrow$$

$$P_1 = P_{o_1} - \rho \frac{u_1^2}{2} \Rightarrow$$

$$Z_1 = Z_{o_1} - \frac{u_1^2}{2g}$$

(1)

$$u_1 = \frac{\dot{m}_1}{\rho_{\text{Lox}} A} = \frac{\dot{m}_1}{\rho_{\text{Lox}} \frac{\pi D_1^2}{4}} = \frac{182}{1140 \times (0.009503)} = 16.8 \frac{\text{m}}{\text{s}}$$

$$\text{From (1)} \quad Z_1 = 23.55 - \frac{(16.8)^2}{2 \times 9.81} = 9.16 \text{ m}$$

(*)

On the other hand, $P_{v, \text{Lox}}$ (vapor pressure of Lox) is

$$= 101.3 \text{ kPa at } 90.3 \text{ K}$$

$$\text{So } Z_{v, \text{Lox}} = \frac{P_{v, \text{Lox}}}{\rho_{\text{Lox}} g} = \frac{101.3 \times 10^3}{1140 \times 9.81} = 9.06 \text{ m}$$

(**)

Compare (*) with (**), $Z_1 > Z_{v, \text{Lox}}$ $9.16 > 9.06$
No cavitation!

(P-2)

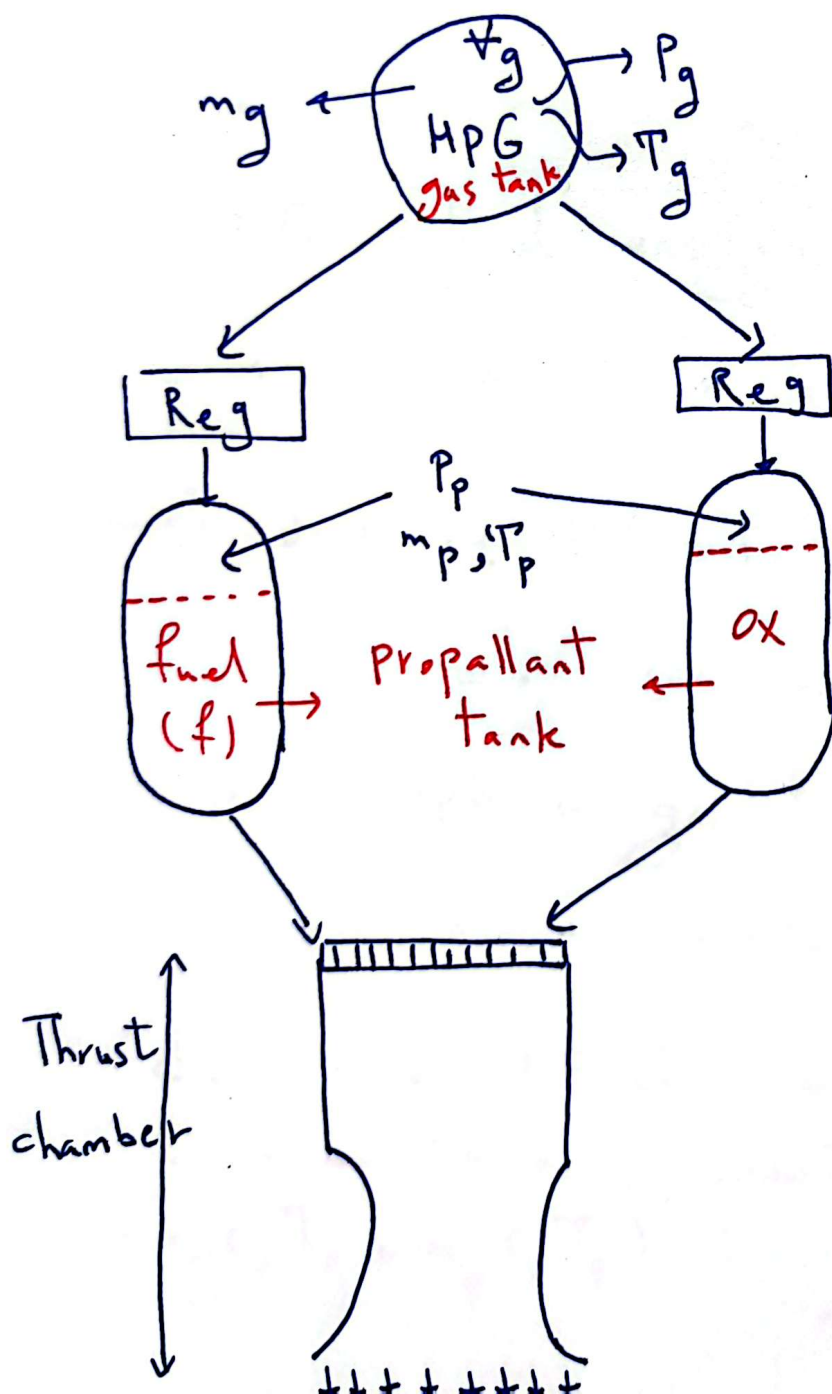
We define "Net positive suction head" for pumps.
 In this example, $NPSH)_{Lox} = \frac{P_0 - P_v}{\rho_{Lox} g} = z_{o_1} - z_v =$

$$23.55 - 9.06 = 14.49 \text{ m}$$

this is the limit above cavitation!

"LRPE Propellant Feed System"

① Gas pressure feed system:



P : pressure of high pressure gas (HPG)

P_g : pressure in gas tank, initially P_{g_0}

$$P_g = P_g(t)$$

P_p : pressure in the propellant tank

m : mass of the HPG

m_g : mass in the gas tank, initially m_{g_0}

$$m_g = m_g(t)$$

m_p : mass in the propellant tank, initially zero

$$m_p = m_p(t)$$

(P-3)

The HPG is used to expel the propellant from their tank to the thrust chamber at the desired pressure, set by the regulator.

Assume adiabatic expansion of the HPG, and apply the first law to the fluid p_g at the instant a volume V_p of the propellant (fuel and oxidizer together) has been delivered to the thrust chamber. The propellant is

" " " " " at pressure P_p .

Initial HPG internal energy = $(m_g c_v T_g)$ $\xrightarrow{\text{initial HPG Temp.}}$

Final " " " = $(m_g c_v T_g + m_p c_v T_p)$

Work done by HPG = $P_p V_p$

Heat = 0 (adiabatic)

T_g : Temp of HPG in gas tank.

T_p : " " " " propellant tank.

$\Delta u = Q - W$ (First law)

$$\underbrace{(m_g c_v T_g + m_p c_v T_p)}_{V_2} - \underbrace{(m_g c_v T_g)}_{V_1} = \underbrace{0}_Q - \underbrace{P_p V_p}_W$$

(P. 4)

Using E.O.S $\Rightarrow pV = mRT \Rightarrow mT = \frac{pV}{R}$ put in

first law: $(c_v \frac{p_g V_g}{R} + c_v \frac{p_p V_p}{R}) - c_v \frac{p_{g_0} V_g}{R} = 0 - p_p V_p$

$$\Rightarrow c_v \frac{p_g V_g}{R} = c_v \frac{p_{g_0} V_g}{R} - c_v \frac{p_p V_p}{R} - p_p V_p$$
$$- \frac{c_p}{R} p_p V_p$$

$$\Rightarrow \text{or: } \boxed{p_g = p_{g_0} - \gamma \frac{V_p}{V_g} p_p} \quad (*)$$

When the pressure of the HPG in the gas tank (p_g) becomes (p_p), no more delivery is possible. By setting $p_g = p_p$ in $(*)$, we can find the minimum required initial pressure of HPG, in the gas tank (p_{g_0}) to deliver volume V_p of propellant to the thrust chamber at pressure p_p .

put $p_g = p_p$ in $(*) \Rightarrow$

$$\boxed{p_p = \overset{\text{given}}{p_{g_0}} - \gamma \frac{V_p}{V_g} p_p}$$

$(V_g \text{ is given})$

$$\text{or } \Rightarrow \boxed{V_g = \frac{\gamma p_p V_p}{\underbrace{p_{g_0} - p_p}_{\text{given}}}} \quad (p-5)$$